



M.Sc. Data Analytics and Technologies

Data Mining and Machine Learning

Student name: Ene, Victor

Student number: 2318217

Name of Instructor: Dr Hewage, Pradeep

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Abstract

The seafood industry plays a vital role in the UK economy, contributing to employment and food security, yet it faces challenges such as fluctuating market demands and regulation constraints. In response, businesses are increasingly adopting approaches to optimise trade and supply chain management.

This report evaluates various analytical tools, data science approaches, and management strategies applicable to the UK seafood dataset. Tools such as R, Python, SQL, Power BI, Tableau, and Apache Spark are assessed based on their suitability for different analytical tasks. Data science methodologies, including exploratory data analysis (EDA), statistical modelling, machine learning and big data analytics, are explored for their effectiveness in deriving meaningful insights. The findings highlight that Python is best suited for forecasting and machine learning, R excels in statistical analysis, SQL is essential for structured data management, Power BI and Tableau are valuable for business intelligence and visualisation. Addressing ethical concerns and adopting efficient data governance strategies will be crucial for leveraging data science in the seafood industry sustainably.

Keywords: Data Science tools, UK Seafood Supply Chain, Machine Learning, Data Management, Business Intelligence, Ethical AI, Data visualisation, Data Governance

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Chapter One

1.0 Introduction

Data can be collected from almost every part of a business, including operations, manufacturing, supply chain management, consumer behaviour, marketing campaign performance, workflow processes and other parts of businesses. Due to the widespread availability of data, there is growing interest in techniques for extracting knowledge and valuable information from it (Provost and Fawcett, 2013). The steady growing reliance on data science and machine learning in industries such as the UK seafood supply chain has led to significant advancements in data driven decision making.

This report explores the application of data science to the UK seafood supply chain dataset by examining relevant research questions and identify appropriate analytical tools, discuss different data science approaches that can be used for data analysis and assess various data science management approaches that can be used for optimising organisation decision making. The report will conduct a literature review on the ethical challenges associated with data mining and machine learning, with focus on bias associated with machine learning, privacy and transparency, and what necessary steps to take in addressing these ethical challenges. Organisations can leverage data mining techniques to get useful information related to seafood production, trade patterns and pricing trends.

1.1 Background

The UK seafood supply chain is a complex network given the international landscape in the trade of fish and fishery products, that involves various industries, including fisheries, processing companies, distributors, retailers, and the final consumers of these products. Ensuring sustainable seafood supplies that will satisfy national demand requires an understanding of national and international demands (Löfstedt *et al.*, 2025).

With the improvements in data science and machine learning, businesses within the seafood supply chain can leverage data analytics to improve supply chain management and optimise trade practices. R can be used for exploratory data analysis, Python can be used for forecasting and machine modelling and powerful visualisation tools like Power BI and Tableau can be used provide business intelligence and descriptive analysis.

Chapter Two

2.0 Research Question

Data science rely on various tools for not so simple tasks like data collection to visualisation of results from complex processing of the results of analysis (Provost and Fawcett, 2013). With the massive amounts of data that are generated over time and emerging technologies various approaches to data science and ways to manage these massive data need to be explored. Handling data will conform to some regulations or concerns (Kimball and Ross, 2013). This report discusses the following research questions after reviewing multiple literatures.

1. What are the data science tools that can be used for data analysis?
2. What are the data science approaches that can be used for data analysis?
3. What different data management approaches can be used by the organisation to manage the data?
4. What are the ethical challenges and ways to mitigate them in data mining and machine learning for seafood supply chain data?

Chapter Three

3.0 Methodology

This report uses a systematic review methodology, which is an approach for finding and collecting most of the existing research on a target subject (Scheerder, van Deursen and van Dijk, 2017). This report reviews and draw insights from related books, academic articles and journals from the University of Greater Manchester library, O'Reilly Learning, Google Scholar, Science Direct, SpringerLink, ResearchGate and IEEE Repositories. The methodology follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) framework, ensuring transparency and quality reporting in a systematic review (Page *et al.*, 2021).

Relevant literature was sourced from ScienceDirect, IEEE Xplore, Google Scholar, and SpringerLink.

3.1 PRISMA Diagram

The inclusion criteria for selecting literature were the most recent books and articles as shown in the diagram below.

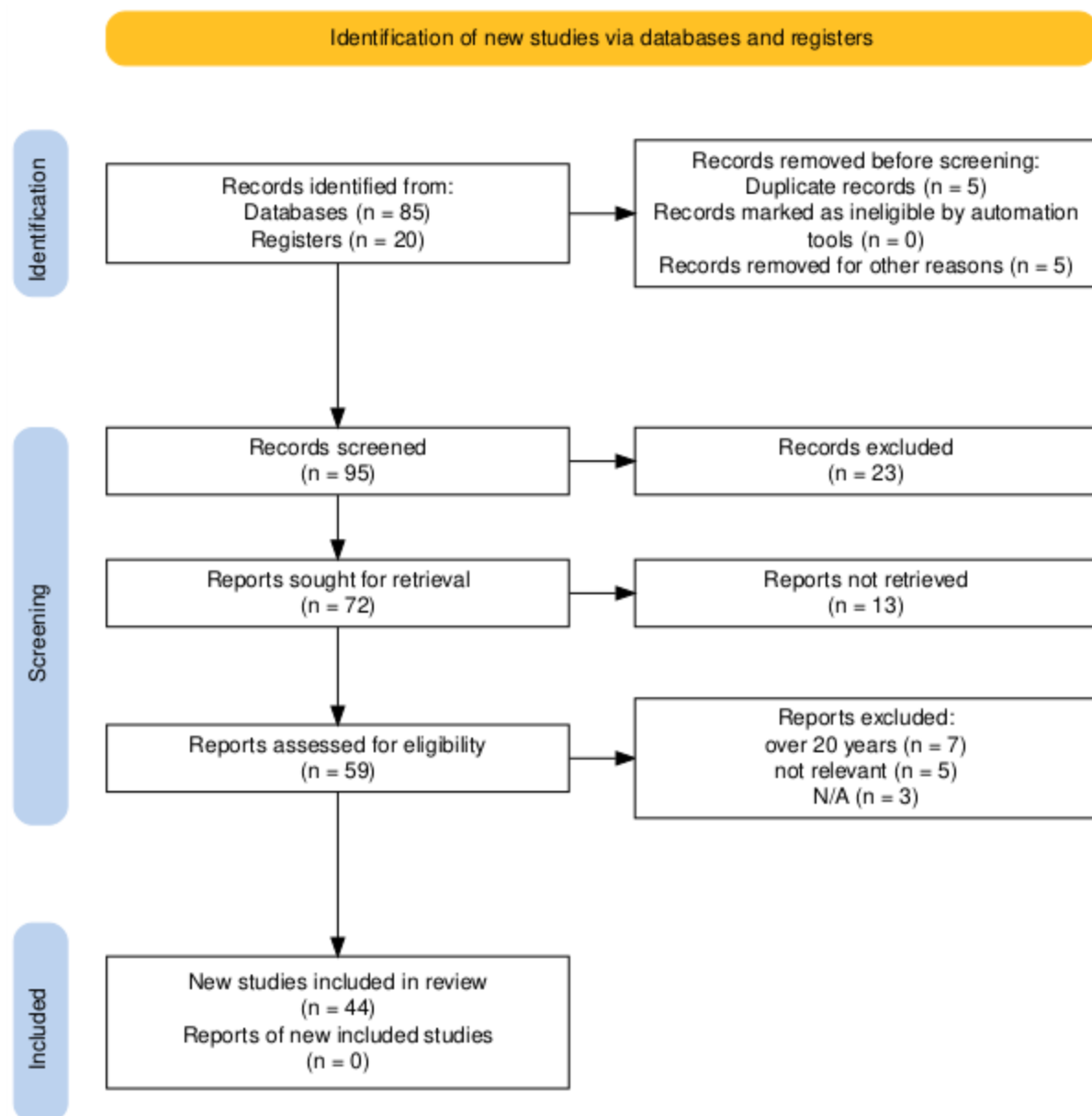


Fig 1.0 Prisma Diagram

3.2 Dataset

The dataset is provided by Her Majesty's Revenue and Customs (HMRC), the dataset consists of 2,691 records and 12 attributes, notable attributes to be used for analysis includes species, trade volume, year, commodity type, and trade classification. The dataset is useful for analysing seafood trade patterns, forecasting demand among others. The dataset provides a detailed view of the UK seafood trade and focuses on the importing, exporting, production and consumption of different seafood commodities over the years from 2009 to 2020. This dataset allows various analysis of UK seafood trade including, patterns in seafood consumption throughout time, as well as predicting seafood import and export volumes, production and consumption volumes and what category of seafood are mostly imported, produced or exported. The image below is a snapshot of the UK supply chain dataset

	A	B	C	D	E	F	G	H	I	J	K	L
1	DataSupplier	DataSet	Commodity	Species	SpeciesType	SACN	Volume	Units	CalendarYear	TemporalFlag	Flag	FlagDescription
2	HMRC	HMRC Overseas	Exports	Abalone	Molluscs	Shellfish	4605000	Grams	2012	Annual	ConV1	ConV1: Con
3	HMRC	HMRC Overseas	Imports	Abalone	Molluscs	Shellfish	10414000	Grams	2012	Annual	ConV1	ConV1: Con
4	HMRC	HMRC Overseas	Exports	Abalone	Molluscs	Shellfish	3931000	Grams	2013	Annual	ConV1	ConV1: Con
5	HMRC	HMRC Overseas	Imports	Abalone	Molluscs	Shellfish	13754000	Grams	2013	Annual	ConV1	ConV1: Con
6	HMRC	HMRC Overseas	Exports	Abalone	Molluscs	Shellfish	200000	Grams	2014	Annual	ConV1	ConV1: Con
7	HMRC	HMRC Overseas	Imports	Abalone	Molluscs	Shellfish	18873000	Grams	2014	Annual	ConV1	ConV1: Con
8	HMRC	HMRC Overseas	Exports	Abalone	Molluscs	Shellfish	1234000	Grams	2015	Annual	ConV1	ConV1: Con
9	HMRC	HMRC Overseas	Imports	Abalone	Molluscs	Shellfish	19264000	Grams	2015	Annual	ConV1	ConV1: Con
10	HMRC	HMRC Overseas	Exports	Abalone	Molluscs	Shellfish	2946000	Grams	2016	Annual	ConV1	ConV1: Con
11	HMRC	HMRC Overseas	Imports	Abalone	Molluscs	Shellfish	13354000	Grams	2016	Annual	ConV1	ConV1: Con
12	HMRC	HMRC Overseas	Exports	Abalone	Molluscs	Shellfish	14156000	Grams	2017	Annual	ConV1	ConV1: Con
13	HMRC	HMRC Overseas	Imports	Abalone	Molluscs	Shellfish	11887000	Grams	2017	Annual	ConV1	ConV1: Con
14	HMRC	HMRC Overseas	Exports	Abalone	Molluscs	Shellfish	20196000	Grams	2018	Annual	ConV1	ConV1: Con
15	HMRC	HMRC Overseas	Imports	Abalone	Molluscs	Shellfish	13235000	Grams	2018	Annual	ConV1	ConV1: Con
16	HMRC	HMRC Overseas	Exports	Abalone	Molluscs	Shellfish	17454000	Grams	2019	Annual	ConV1	ConV1: Con
17	HMRC	HMRC Overseas	Imports	Abalone	Molluscs	Shellfish	7861000	Grams	2019	Annual	ConV1	ConV1: Con
18	HMRC	HMRC Overseas	Exports	Abalone	Molluscs	Shellfish	8318000	Grams	2020	Annual	ConV1	ConV1: Con

Fig 2.0 UK seafood supply chain <https://reshare.ukdataservice.ac.uk/856955/1/856955 bundle.zip>

Chapter Four

4.0 Data Science Tools

Data science tools encompass a range of software programming environments and frameworks that are designed to assist in data collection, processing, analysis, visualisation, and predictive modelling. These tools empower data scientists to extract meaningful insights from both structured and unstructured data and manage them effectively to help drive informed decision making (Provost and Fawcett, 2013).

4.1 R

R is an open source programming language designed for statistical computing and visualisation (McGrath, 2018). R is widely used for its ability to produce high quality plots and graphics, making it a preferred tool for statistical modelling (R Core Team, 2023). R was developed in the early 1990s and since then R has been among the most important tools for data analysis and visualisation (Zamora Saiz *et al.*, 2020). With packages like ggplot2, tidyr, readr, dplyr and other packages that extend the capabilities of R making it easy to use (Wickham, Cetinkaya-Rundel and Grolemund, 2023).

4.2 Python

Since appearing in 1991, the python programming language has been widely used among other interpreted programming languages and over the years it has become one of the most important languages for machine learning and data analysis (McKinney, 2022). Although Python was not originally designed for analysing data or scientific computing, Python will unavoidably be compared to other widely used open source and commercial programming languages and tools

for data analysis, interactive computing, and data visualisation of large datasets, this is as a result of third-party packages which includes Pandas, NumPy, SciPy, Matplotlib, IPython, Skikit-Learn and other packages that ease data cleaning, analysis, visualisation, and machine learning (Jake VanderPlas, 2022).

4.3 SQL (Structured Query Language)

SQL was created in the early 1970s to be a standard method for accessing and manipulating relational data, over the years many relational database developers have developed languages to access and manipulate relational data, only a few have been universally accepted and simple to learn (Kline, Obe and Hsu, 2022). Different relational database management systems have a slight difference in how they implement SQL, some of these relational database management systems are MySQL, Microsoft SQL Server, Oracle Database, PostgreSQL among others, useful for querying and managing structured data in databases (Zhao, 2021).

4.4 Power BI

Power BI is a powerful business intelligence and analytics tool developed by Microsoft, enabling analysts and developers to access various data sources, structure data models and create interactive visual reports and dashboards (Knight *et al.*, 2018). Power query, a data wrangling tool is built into Power BI as well as the DAX expression and query language and Vertipaq engine that performs complex calculations efficiently and other visualisations (Wade, 2020).

4.5 Tableau

Tableau was developed to enable anyone, regardless of their technical proficiency, to create a customised dashboard and evaluate their data. Tableau can access data from various data sources,

simple files, Hadoop clusters, cloud services and databases and then use its interface to display results in various type of charts, visualisations in Tableau are created in the form of worksheets, dashboards and storyboard (Loth, 2019).

4.6 Apache Spark

Apache Spark is designed to support a wide range of data analytics tasks from SQL queries to large-scale data processing and machine learning with the same computational engine and a standardised set of APIs (Chambers and Zahari, 2018).

Chapter Five

5.0 Data Science Approaches

Data science approaches use mathematical modelling, machine learning, and statistical techniques to solve problems across industries some of them are:

5.1 Exploratory Data Analysis (EDA)

In Exploratory Data Analysis we try to understand what is contained in the data. Exploratory Data Analysis is usually the first step in any data science process, where analysts investigate datasets to uncover patterns, relationships, trends and outliers before applying various formal modelling techniques (Mukhiya and Ahmed, 2020). This approach involves summarising data using descriptive statistics, visualising data and identifying missing or inconsistent values, in a data driven organisation, it can be considered to be the most important step and a lot of time should be allocated for proper EDA (Datar and Garg, 2019). Performing EDA on the UK seafood supply chain dataset would include:

- I. Loading the dataset into our preferred tool
- II. Performing data cleaning and preprocessing.
- III. Conducting univariate, bivariate, and multivariate analyses.
- IV. Detecting and handling outliers.
- V. Performing feature transformation if needed.
- VI. Documenting findings and observations.

5.2 Statistical Approaches

Statistical approaches are the systematic techniques used for designing experiments, collecting data, building models and making predictions of numerical data (Montgomery, Peck and Vining, 2021). By using mathematical and computer techniques to find patterns, connections, and probabilities in data, statistical methods are used to extract valuable insights (Anderson *et al.*, 2017). Some statistical techniques are:

- I. Descriptive Statistics: This is the formation of simple data summaries and basic characteristics, they could be numerical to clearly understand or visualisations, examples are mean, median, mode, standard deviations, graphs and histograms (Mukhiya and Ahmed, 2020).
- II. Inferential statistics: is a field of statistics that enables us to draw conclusions, make predictions, or gather insights about an entire population using a sample of data (Montgomery and Runger, 2018).
- III. Hypothesis Testing: Hypothesis testing is used to test if a conclusion is valid for specific datasets by comparing the data against a certain assumption. Using t-tests, chi-square tests, and ANOVA to validate assumptions about different groups (McClave, Benson and Sincich, 2022).
- IV. Regression Models: Applying linear regression for continuous dependent variables and logistic regression for binary classification tasks (Hoffmann, 2022).
- V. Time-Series Analysis: A time series is an ordered sequence of values recorded over an equally spaced period of time, some examples are moving averages, Integrated

series, smoothing methods, and auto regressive moving average series to analyse trends and seasonal patterns in data over time (Shumway and Stoffer, 2025).

5.3 Machine Learning (ML)

Machine learning focuses on developing algorithms that learn patterns from data to make decisions under doubts, ambiguity or indecision and also make predictions (Murphy, 2023).

Machine learning methods are classified into:

- I. Supervised Learning: Involves training models on labelled data examples are classification and regression (Murphy, 2023). Algorithms for supervised learning include decision trees, support vector machines, and neural networks.
- II. Unsupervised Learning: is a type of machine learning that involves training models when labels are unavailable, this means that it identifies patterns without already defined labels examples are clustering, anomaly detection, visualisation and dimensionality reduction, association rule learning, with algorithms including k-means clustering, isolation forest, eclat and principal component analysis (Géron, 2023).
- III. Reinforcement Learning: Reinforcement learning focuses on training and learning through rewards and penalties, actions which yield the best results are discovered by trying and getting errors (Sutton and Barto, 2020).

5.4 Big Data Analytics

Big Data Analytics is the processing of massive datasets and extracting insights from these massive datasets that exceed the capabilities of usual databases (Hashem *et al.*, 2015). Big Data Analytics

enables real time decision making, detecting anomalies and predictive modelling at a tremendous rate with the advancements in cloud computing and artificial intelligence technologies (Hashem *et al.*, 2015). This approach is useful in industries like healthcare, security, finance, social media, virtualised resources and any industry where very large amounts of real time data are generated (Marr, 2015). Some big data technologies are:

- I. Hadoop
- II. Apache Spark
- III. MongoDB
- IV. Flink
- V. Google BigQuery
- VI. Amazon Redshift

Chapter Six

6.0 Data Science Management Approaches

Effective data science management approach is useful in for extracting meaningful insights, ensuring ethical data handling, and optimising decision making. in the UK seafood supply chain, organisations managing this dataset can adopt various data science management approaches, including but not limited to

6.1 Cross Industry Standard Process for Data Mining (CRISP-DM)

CRISP-DM is one of the very popular approaches for managing data science projects, it provides a structured, six phase approach as shown in fig 3. below. It ensures that projects follows a logical sequence, from understanding the objectives of the business to deploying models.

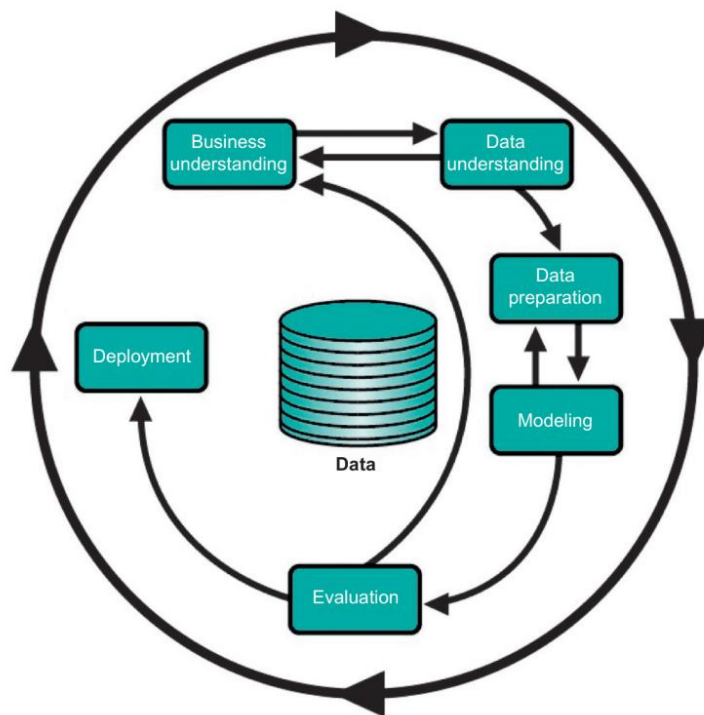


Fig 3. Phases of the CRISP-DM process (Nisbet, Miner and McCormick, 2024).

This approach can be applied in the UK seafood supply chain dataset by

- I. Defining the business objectives such as predicting demand trends and patterns in consumption.
- II. Understanding the UK seafood supply chain dataset.
- III. Prepare the data by handling missing values, outliers, using standard units.
- IV. Model the data using modelling tools like ARIMA to predict trends.
- V. Evaluating data against past records.
- VI. Deploy the model and visualise the results such as creating a dashboard to monitor real time trade.

6.2 Agile Data Science Management

Agile methodologies were originally developed for software development, have been adapted to data science to foster flexibility, collaboration, and iterative progress. The Agile approach breaks down projects into smaller parts, increasing little by little, allowing teams to refine models continuously based on the feedback from stakeholders. There are twelve principles of agile software which are:

- I. Customer satisfaction as a priority through early and continuous delivery of valuable software such as ensuring stakeholders in the seafood industry get early updates on price and demand trends (Stellman and Greene, 2015).
- II. Welcome changing requirements, Agile processes allow for changes in conditions where the seafood chain is affected by regulations or weather conditions (Stellman and Greene, 2015).

- III. Agile enables delivering working software frequently and at a very short timescale, like displaying real time seafood prices (Stellman and Greene, 2015).
- IV. The most effective way to communicate within a development team is through direct, face-to-face discussions (Stellman and Greene, 2015).
- V. Developers in the industry and business stakeholders should collaborate daily throughout the project's duration (Stellman and Greene, 2015).
- VI. Projects should be built around motivated individuals, offering them the necessary support and trusting them to succeed (Stellman and Greene, 2015).
- VII. The primary indicator of progress is delivering working software (Stellman and Greene, 2015).
- VIII. Agile fosters sustainable development, ensuring that sponsors, developers and users can maintain a steady workflow indefinitely (Stellman and Greene, 2015).
- IX. Focusing on technical excellence and well-structured design enhances adaptability (Stellman and Greene, 2015).
- X. Simplicity, which involves maximising efficiency by minimising unnecessary work, is a key principle (Stellman and Greene, 2015).
- XI. The most effective architectures, requirements, and designs emerge from self-driven, collaborative teams (Stellman and Greene, 2015).
- XII. Teams should regularly assess their processes and refine their approach for continuous improvement (Stellman and Greene, 2015).

6.3 Data Science Operations Management (DataOps)

Data Science Operations Management is a very recent management approach that focuses on automating, monitoring, and optimising the data pipeline lifecycle. DataOps steps in providing a structured, flexible framework for data operations and enhance data agility in the organisation, implementing DataOps effectively is crucial for large organisations that want to have an advantage over others with optimised data management and analytics (Gift and Deza, 2021). Key practices include:

I. Continuous Integration

This is an automated process that tests and improve the quality of a software project (Gift and Deza, 2021).

II. Continuous Delivery

This involves automatically deploying code to a new environment. Often, continuous delivery workflows incorporate Infrastructure as code to ensure consistency and speed (Gift and Deza, 2021).

III. Microservices

A microservice is a small, self-contained software component with minimal dependencies. Frameworks like Flask can be used to build a prediction endpoint or other functionalities, and these services can be deployed using platforms such as AWS Lambda or Azure App Services (Gift and Deza, 2021).

IV. Infrastructure as Code

Infrastructure as code stores infrastructure configurations in a version-controlled repository and deploys changes automatically. This ensures

consistent environments without manual setup platform such as AWS CloudFormation and Terraform are used (Gift and Deza, 2021).

V. Monitoring and Instrumentation

These practices help organisations identify and resolve issues quickly, it involves collecting performance data from New Relic or Stackdriver to guide improvements (Gift and Deza, 2021).

VI. Effective Technical Communication

This involves concise, repeatable communication strategies example documentation and automation streamline collaboration. Using AutoML for initial prototypes can reveal project feasibility in the early stages (Gift and Deza, 2021).

VII. Effective Technical Project Management

This involves breaking the project into smaller tasks and focusing on little increasing wins to boost steady progress and limit risk (Gift and Deza, 2021).

6.4 Machine Learning Operations management (MLOps)

MLOps is a new emerging technique for bringing Machine Learning and Software Development together by combining domains of different kind (Raj, 2021). MLOps combines machine learning, data engineering and DevOps with the goal of reliably and efficiently building, deploying, and maintaining ML systems in production.

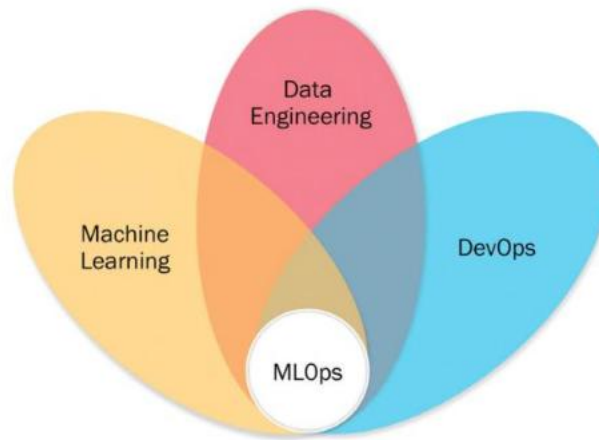


Fig. 4 MLOps Diagram

MLOps fuses Machine Learning, DevOps, and data engineering, to have reliably data and efficiently building, deploying, and maintaining machine learning systems.

6.5 Lean Data Science Management

It focuses on cutting unnecessary steps, making processes more efficient, and quickly delivering real business value. Inspired by lean manufacturing and startup principles, this approach ensures that data-driven insights are generated with minimal waste.

Chapter Seven

7.0 Ethical Challenges in Data Mining and Machine Learning

The advancement of data mining and machine learning has significantly transformed various industries, including the seafood supply chain. While these technologies ensure getting valuable insights in supply and demand, pricing strategies and sustainability efforts, they also present significant ethical challenges (Mukherjee *et al.*, 2023). In the UK Seafood Supply Chain dataset, key ethical concerns include bias, privacy and transparency. Addressing these challenges is crucial for ensuring fairness, trust and regulatory compliance in making data-driven decisions (Gill and Hall, 2019).

The UK seafood supply chain faces a disconnect between the harvesting sector and downstream operations. Harvesters focus on the opportunities in fishing, making sure that prices are secure and constraints in regulation, while processors and retailers are focused on product quality and market access (Symes and Phillipson, 2019).

Traditionally, companies in the seafood supply chain have operated independently, focusing on their own social, economic, and environmental improvements. Research suggests that collaborations across the entire supply chain can improve product quality and profit without increasing the use of resources. To drive industry growth, these companies must improve communication and coordinate strategies with suppliers and customers (Denham *et al.*, 2015). While the shellfish sector for example is valued at £339 million and heavily reliant on EU exports it faced uncertainty in trade negotiations. UK fishermen advocate for separate discussions on

fishing rights and tariffs, whereas EU counterparts seek to link them in reciprocal negotiations (Symes and Phillipson, 2019).

7.1 Bias in Data and Algorithms

Bias in machine learning occurs when a model systematically favours a certain group due to incomplete and misrepresented data or an imbalanced criterion. Within the UK seafood industry, biases may occur in pricing predictions when some prices are not properly mapped or in supplier recommendations.

7.1.1 Types of Bias in the Dataset

- I. Selection Bias: If the dataset does not properly represent the proportion of certain seafood categories over another, or misinterpret the actual population of certain categories, then analytical models may generate skewed pricing forecasts (Mukherjee *et al.*, 2023).
- II. Algorithmic Bias: Training a predictive model on historical seafood trade data may be affected by existing market disparities, algorithm bias is caused by the algorithm it could be regulations, some functions used in optimising or other causes (Pagano *et al.*, 2023).
- III. Confirmation Bias: Analysts may focus on data that aligns with preconceived industry assumptions, leading to misleading conclusions for example a company may believe that certain seafood suppliers always deliver late and might focus only on instances of delayed shipments, ignoring cases where the same suppliers met or exceeded delivery expectations.

7.1.2 Mitigation Strategies

- I. Ensuring dataset diversity by incorporating data from small and large suppliers, different geographic regions, and varied fishing methods (Mukherjee *et al.*, 2023).
- II. Implementing fairness algorithms to prevent discrimination in pricing or market access predictions (Mukherjee *et al.*, 2023).
- III. Conducting regular bias audits and fairness tests on machine learning models (Gill and Hall, 2019).

7.2 Privacy and Data Protection

Data mining often involves handling sensitive business and personal information, including seafood trade records, pricing agreements, and supplier contracts, most of these information can affect businesses and even cause harm If not properly managed as the data may be vulnerable to breaches or misuse (Tamò-Larrieux, 2018).

7.2.1 Privacy Concerns in the Dataset

- I. Confidentiality of Supplier Data: Unauthorised access to trade agreements and financial records could lead to unfair competitive advantage (Tamò-Larrieux, 2018).
- II. Consumer Data Protection: If the dataset contains purchase trends or other personal information, mishandling such data could violate the UK General Data Protection Regulation (UK GDPR) (Tamò-Larrieux, 2018).
- III. Black box phenomenon: This occurs when organisations analyse datasets to get useful insights by observing patterns on their behavioural pattern to predict future behaviour, methods used to obtain these insights are kept from the data subjects and this becomes

an issue when the data the organisation has observed is wrong or has inconsistent values (Tamò-Larrieux, 2018).

7.2.2 Mitigation Strategies

- I. Applying different privacy techniques to ensure data anonymity (Tamò-Larrieux, 2018).
- II. Implementing access controls and encryption for sensitive data (Tamò-Larrieux, 2018).
- III. Adhering to UK GDPR and other regulatory standards to ensure ethical data handling (Pagano *et al.*, 2023).

7.3 Transparency

Transparency is ensuring that regulators and consumers understand how predictions and recommendations are generated. A lack of transparency can lead non-compliance with industry regulations.

7.3.1 Transparency Issues

- I. Opaque Pricing Models: If machine learning algorithms determine seafood pricing without explanation, stakeholders may suspect manipulation.
- II. Regulatory Challenges: Authorities may require AI-based decisions to be explainable, especially when compliance with fishing quotas and sustainability laws is at stake.
- III. Stakeholder Distrust: Fishermen, suppliers, and retailers may resist AI-driven insights if they cannot interpret the decision-making logic.

7.3.2 Mitigation Strategies

- I. Using interpretable machine learning models, such as decision trees (Gill and Hall, 2019).
- II. Providing detailed documentation and transparency reports on AI generated decision-making processes (Caton and Haas, 2024).
- III. Ensuring to adhere to regulations by maintaining model explainability for compliance checking and auditing (Tamò-Larrieux, 2018).

The UK Seafood Supply Chain dataset presents significant ethical challenges in bias, privacy, and transparency. Bias in data and algorithms can cause market inequalities, privacy risks may compromise sensitive trade information, and lack of transparency can undermine trust in decisions generated using artificial intelligence. Addressing these ethical concerns requires different approaches, including auditing for bias, privacy preserving techniques and using artificial intelligence frameworks that can be explained. Responsible use of data mining and machine learning is essential for a fair, safe, transparent, and sustainable seafood industry.

8.0 Discussion

The different data science tools have situations where they are best to use and situation that they are not appropriate to use, below is a table that shows the strengths, weaknesses and best cases of the data science tools discussed in this report

Tool	Strength	Weakness	Best used in
R	Excellent for statistical analysis and visualisation, suitable for small datasets	Steeper learning curve, slower for handling large-scale data	Statistical modelling, trend analysis on trade volumes
Python	Versatile, widely used for machine learning, data analysis, and visualisation	Can be slower for big data processing compared to Spark	Machine learning, data visualisation, and predictive analytics
SQL	Efficient for querying and managing structured databases, ideal for relational data	Limited for unstructured data and complex analytics	Database management, efficient querying of seafood trade records

Power BI	Great for business intelligence and interactive reporting	Less flexible for advanced machine learning tasks	Business intelligence, tracking seafood trade trends
Tableau	User-friendly dashboards and visual analytics	Not ideal for large-scale machine learning tasks	Interactive dashboards for monitoring seafood trade statistics
Apache Spark	Handles big data and distributed computing efficiently	Requires more infrastructure and computational resources	Processing large datasets, distributed computing for trade patterns

Table 1.0 Comparing data science tools

9.0 Conclusion and Recommendations

The increasing reliance on data science in the UK seafood supply chain has introduced significant opportunities for data-driven decision making. Through the application of data analysis tools such as R, Python, SQL, Power BI, businesses are able to optimise production, trade, and supply chain operations. Various data science approaches, including exploratory data analysis (EDA), statistical modeling and machine learning algorithms enable organisations to uncover patterns, detect trends, and improve forecasting.

However, despite the benefits of data science applications, the seafood supply chain faces critical ethical challenges, including bias in machine learning models, data privacy concerns, and transparency issues, to address these challenges requires ethical frameworks, fairness algorithms and regulatory compliance to ensure the ethical use of data.

Effective data science management approaches such as CRISP-DM, Agile Data Science Management, DataOps, MLOps, and Lean Data Science Management play a crucial role in optimising data processes and ensuring the reliability of analytical outputs. By implementing best practices in data governance, and ethical AI practices, organisations within the UK seafood supply chain can achieve enhanced efficiency. Moving forward, continuous research, innovation, and ethical considerations will be essential to fully harness the potential of data science in the seafood sector.

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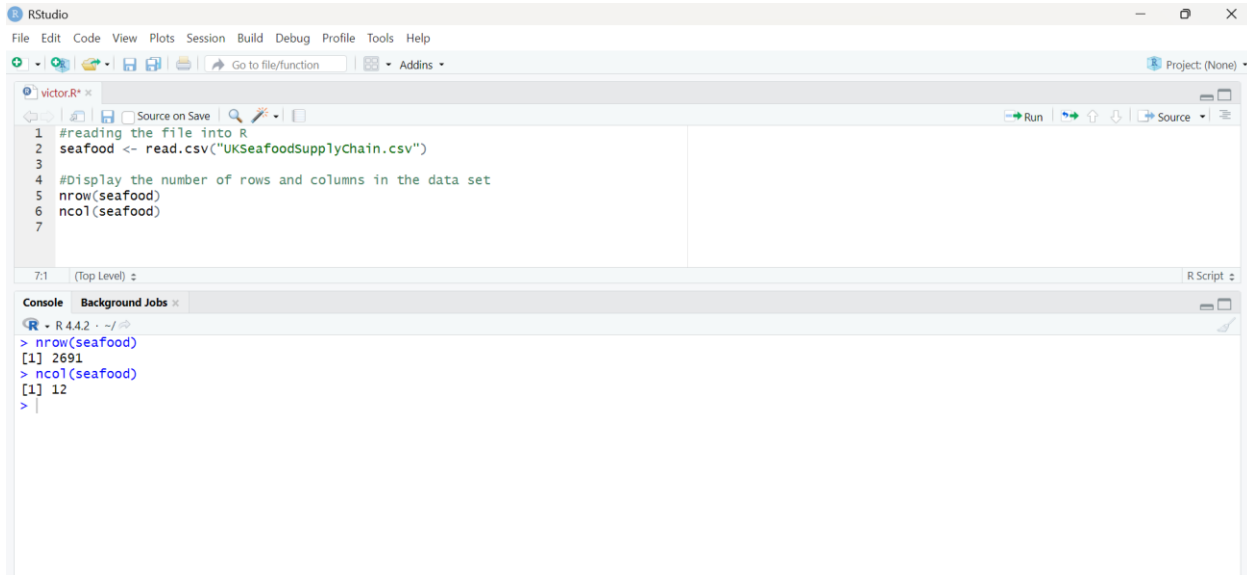
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Appendix



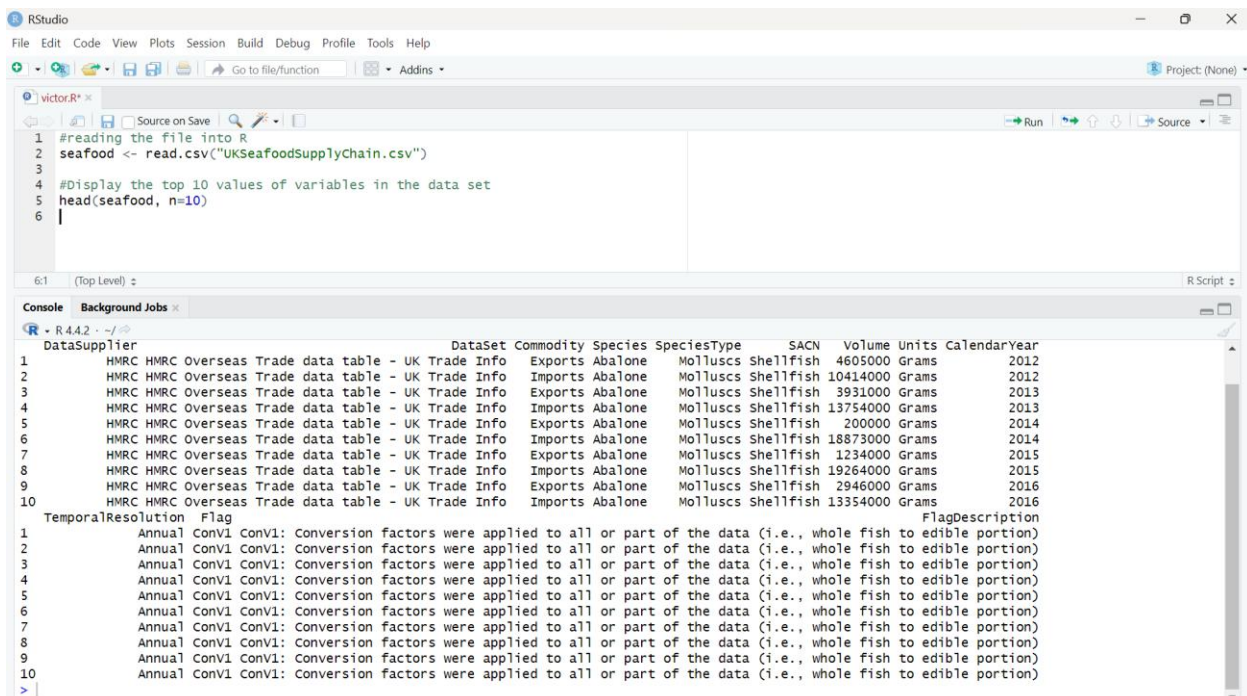
The screenshot shows the RStudio interface. The script editor contains the following R code:

```
1 #reading the file into R
2 seafood <- read.csv("UKSeafoodSupplyChain.csv")
3
4 #Display the number of rows and columns in the data set
5 nrow(seafood)
6 ncol(seafood)
7
```

The console output shows the results of the commands:

```
> nrow(seafood)
[1] 2691
> ncol(seafood)
[1] 12
>
```

Fig. 5 Showing an R code that displays the number of rows and column in the dataset



The screenshot shows the RStudio interface. The script editor contains the following R code:

```
1 #reading the file into R
2 seafood <- read.csv("UKSeafoodSupplyChain.csv")
3
4 #Display the top 10 values of variables in the data set
5 head(seafood, n=10)
6
```

The console output shows the first 10 rows of the dataset:

	DataSupplier	TemporalResolution	Flag	DataSet	Commodity	Species	SpeciesType	SACN	Volume	Units	CalendarYear
1	HMRC	Annual	Conv1	Overseas Trade data table - UK Trade Info	Exports	Abalone	Molluscs Shellfish	4605000	Grams		2012
2	HMRC	Annual	Conv1	Overseas Trade data table - UK Trade Info	Imports	Abalone	Molluscs Shellfish	10414000	Grams		2012
3	HMRC	Annual	Conv1	Overseas Trade data table - UK Trade Info	Exports	Abalone	Molluscs Shellfish	3931000	Grams		2013
4	HMRC	Annual	Conv1	Overseas Trade data table - UK Trade Info	Imports	Abalone	Molluscs Shellfish	13754000	Grams		2013
5	HMRC	Annual	Conv1	Overseas Trade data table - UK Trade Info	Exports	Abalone	Molluscs Shellfish	200000	Grams		2014
6	HMRC	Annual	Conv1	Overseas Trade data table - UK Trade Info	Imports	Abalone	Molluscs Shellfish	18873000	Grams		2014
7	HMRC	Annual	Conv1	Overseas Trade data table - UK Trade Info	Exports	Abalone	Molluscs Shellfish	1234000	Grams		2015
8	HMRC	Annual	Conv1	Overseas Trade data table - UK Trade Info	Imports	Abalone	Molluscs Shellfish	19264000	Grams		2015
9	HMRC	Annual	Conv1	Overseas Trade data table - UK Trade Info	Exports	Abalone	Molluscs Shellfish	2946000	Grams		2016
10	HMRC	Annual	Conv1	Overseas Trade data table - UK Trade Info	Imports	Abalone	Molluscs Shellfish	13354000	Grams		2016

Below the table, the 'Flag' column is expanded, showing the following descriptions for the first 10 rows:

```
1 Conv1 Conv1: Conversion factors were applied to all or part of the data (i.e., whole fish to edible portion)
2 Conv1 Conv1: Conversion factors were applied to all or part of the data (i.e., whole fish to edible portion)
3 Conv1 Conv1: Conversion factors were applied to all or part of the data (i.e., whole fish to edible portion)
4 Conv1 Conv1: Conversion factors were applied to all or part of the data (i.e., whole fish to edible portion)
5 Conv1 Conv1: Conversion factors were applied to all or part of the data (i.e., whole fish to edible portion)
6 Conv1 Conv1: Conversion factors were applied to all or part of the data (i.e., whole fish to edible portion)
7 Conv1 Conv1: Conversion factors were applied to all or part of the data (i.e., whole fish to edible portion)
8 Conv1 Conv1: Conversion factors were applied to all or part of the data (i.e., whole fish to edible portion)
9 Conv1 Conv1: Conversion factors were applied to all or part of the data (i.e., whole fish to edible portion)
10 Conv1 Conv1: Conversion factors were applied to all or part of the data (i.e., whole fish to edible portion)
```

Fig. 6 showing the top 10 values on the variables in the dataset

```

1 #reading the file into R
2 seafood <- read.csv("UKSeafoodSupplyChain.csv")
3
4 #Display the top 10 values of variables in the data set
5 head(seafood, n=10)
6

```

DataSupplier	Dataset	Commodity	Species	SpeciesType	SACN	Volume	Units	CalendarYear
HMRC	HMRC Overseas Trade data table - UK Trade Info	Exports Abalone	Molluscs	Shellfish	4605000	Grams	2012	
HMRC	HMRC Overseas Trade data table - UK Trade Info	Imports Abalone	Molluscs	Shellfish	10414000	Grams	2012	
HMRC	HMRC Overseas Trade data table - UK Trade Info	Exports Abalone	Molluscs	Shellfish	3931000	Grams	2013	
HMRC	HMRC Overseas Trade data table - UK Trade Info	Imports Abalone	Molluscs	Shellfish	13754000	Grams	2013	
HMRC	HMRC Overseas Trade data table - UK Trade Info	Exports Abalone	Molluscs	Shellfish	200000	Grams	2014	
HMRC	HMRC Overseas Trade data table - UK Trade Info	Imports Abalone	Molluscs	Shellfish	18873000	Grams	2014	
HMRC	HMRC Overseas Trade data table - UK Trade Info	Exports Abalone	Molluscs	Shellfish	1234000	Grams	2015	
HMRC	HMRC Overseas Trade data table - UK Trade Info	Imports Abalone	Molluscs	Shellfish	19264000	Grams	2015	
HMRC	HMRC Overseas Trade data table - UK Trade Info	Exports Abalone	Molluscs	Shellfish	2946000	Grams	2016	
HMRC	HMRC Overseas Trade data table - UK Trade Info	Imports Abalone	Molluscs	Shellfish	13354000	Grams	2016	

Fig. 7 showing the last 10 values of the variables in the dataset

```

1 #reading the file into R
2 seafood <- read.csv("UKSeafoodSupplyChain.csv")
3 #Display the structure of the data set
4 str(seafood)
5
6

```

```

> str(seafood)
'data.frame': 2691 obs. of 12 variables:
 $ DataSupplier : chr "HMRC" "HMRC" "HMRC" "HMRC" ...
 $ Dataset : chr "HMRC Overseas Trade data table - UK Trade Info" "HMRC Overseas Trade data table - UK Trade Info" "HMRC Overseas Trade data table - UK Trade Info" ...
 $ Commodity : chr "Exports" "Imports" "Exports" "Imports" ...
 $ Species : chr "Abalone" "Abalone" "Abalone" "Abalone" ...
 $ SpeciesType : chr "Molluscs" "Molluscs" "Molluscs" "Molluscs" ...
 $ SACN : chr "Shellfish" "Shellfish" "Shellfish" "Shellfish" ...
 $ Volume : num 4605000 10414000 3931000 13754000 200000 ...
 $ Units : chr "Grams" "Grams" "Grams" "Grams" ...
 $ CalendarYear : int 2012 2012 2013 2013 2014 2015 2015 2016 2016 ...
 $ TemporalResolution : chr "Annual" "Annual" "Annual" "Annual" ...
 $ Flag : chr "ConV1" "ConV1" "ConV1" "ConV1" ...
 $ FlagDescription : chr "ConV1: Conversion factors were applied to all or part of the data (i.e., whole fish to edible portion)" "ConV1: Conversion factors were applied to all or part of the data (i.e., whole fish to edible portion)" "ConV1: Conversion factors were applied to all or part of the data (i.e., whole fish to edible portion)" ...

```

Fig 8 showing the structure of the dataset